

Haotian Lin

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Personal Website: <https://darthutopian.github.io>

EDUCATION

Carnegie Mellon University

M.S. in Robotics

Pittsburgh

Aug. 2024– now

- GPA: 4.0/4.0
- Related Courses: Advanced Machine Learning (A), Optimal Control & RL (A), Introduction to Robot Learning (A)

Tsinghua University

B.S. in Mechanics & B.E. in Vehicle Engineering

Beijing

Sep. 2020 – Jun. 2024

University of California, Berkeley

Visiting Research Intern (Undergraduate)

Beijing

Jun. 2023 – Sep. 2024

RESEARCH INTERESTS

I am broadly interested in the intersection of reinforcement learning and robotics foundation models. Specifically, I am interested in scalable RL algorithms and improving robotic foundation models with on-the-fly experience. I broadly explore novel algorithms, architectures, and pre-training/post-training paradigms that align with my goal. Eventually, I aim to build scalable, self-improving, generalist robotic agents.

RESEARCH EXPERIENCE

LeCAR Lab & Auton Lab (Carnegie Mellon University)

Pittsburgh, PA

Graduate Research Assistant (MSR)

Advisor: [Guanya Shi](mailto:guanyas@andrew.cmu.edu) (guanyas@andrew.cmu.edu), [Jeff Schneider](mailto:jeff4@andrew.cmu.edu) (jeff4@andrew.cmu.edu)

Multi-task RL for Robotics in Parallel Simulation

6/2025–Now

- Study transferable skill and task representation learning in multi-task learning, especially for on-policy RL algorithms with massive parallel simulations.
- Contributed to HumanoidVerse, a sim2real RL infra. Implement MPPI controller and initial codebase for dexterous manipulation task.

Self-improving Foundation Models via Residual RL

5/2025 – 10/2025

- Building data-efficient RL pipeline for VLA in JAX for simulation benchmarks (LIBERO, SimplerEnv), and real-world tasks. supports data collection through different teleoperation approaches and self-play, integrates SFT and evaluation pipeline.
- Develop PLD, a VLA post-training pipeline characterized by a bootstrapped data flywheel, minimal human supervision, and real-world deployability.

Improving Model-Based RL for High-Dimensional Continuous Control

10/2024 – 3/2024

- Study structural policy mismatch and value overestimation in plan-based MBRL at high-dimensional action space. Provide theoretical groundings for value approximation error.
- Propose a minimalist fix to mitigate the issue for TD-MPC family algorithms and boost performance across a wide variety of continuous control tasks, e.g., humanoid locomotion.

Mechanical System Control Lab (University of California, Berkeley)

Berkeley, CA

Research Assistant

Advisor: [Masayoshi Tomizuka](mailto:tomizuka@berkeley.edu) (tomizuka@berkeley.edu)

Diffusion Model for Multi-Agent Pedestrian Trajectory Prediction

10/2023 – 4/2024

- Leverage latent diffusion to model complex social interaction between agents. Propose a novel trajectory prediction scheme that forms trajectory prediction as unified inverse problem and solves through posterior sampling.
- Contribute to conference paper admitted to IROS 2024.

Prediction Benchmark

7/2023 – 9/2023

- Evaluate prediction models' ability in interaction modeling using Shapley-Value-based contribution assignment metrics, bridging the gap between concept interpretation and performance.

THICV (Tsinghua University)

Beijing

Research Assistant

Advisor: Jianqiang Wang (wjqlws@tsinghua.edu.cn) , Shaobing Xu (shaobxu@tsinghua.edu.cn)

GNN-based prediction framework for generalizable scene understanding

10/2022 – 6/2023

- Designing social interaction aware backbone leveraging prior knowledge for autonomous driving, adaptable to downstream tasks including trajectory prediction and intention detection. Designing conditional diffusion model-based traffic simulator that integrates natural language commands.
- Contribute to conference papers admitted to IEEE IV 2023 and IROS 2025.

iDLab (Tsinghua University)

Beijing

Research Assistant

(Students Research Training Program)

Advisor: Shengbo Li (lishbo@tsinghua.edu.cn)

DRL Algorithm Development for Integrated Decision-Making & Control

10/2022 – 6/2023

- Testing and verification of General Optimal control Problems Solver (GOPS) RL toolkit.
- Develop offline-to-online RL algorithms for diffusion policies using Q-score alignment.

SELECTED PUBLICATIONS

- Wenli Xiao*, **Haotian Lin***, Haoru Xue, Tairan He, Zhengyi Luo, Yuqi Xie, Fengyuan Hu, Andy Peng, Guanya Shi, Jim Fan, Yuke Zhu. "Self-Improving Vision-Language-Action Models with Data Generation via Residual RL" (In Submission)
- **Haotian Lin**, Pengcheng Wang, Jeff Schneider, Guanya Shi. "TD-M (PC) $\mathcal{S}^2$: Improving Temporal Difference MPC Through Policy Constraint". arXiv preprint arXiv:2502.03550. (In Submission)
- Wang Yixiao, Mingxiao Huo, Zhixuan Liang, Yushi Du, Lingfeng Sun, **Haotian Lin**, Jinghuan Shang et al. "Ver: Vision expert transformer for robot learning via foundation distillation and dynamic routing." arXiv preprint arXiv:2510.05213. (In Submission)
- **Haotian Lin**, Yixiao Wang, Mingxiao Huo, Chensheng Peng, Zhiyuan Liu, Masayoshi Tomizuka. "Joint Pedestrian Trajectory Prediction through Posterior Sampling". 2024 IEEE/RSJ International Conference on Intelligent Robots and Systems.
- Zhiyuan Liu, Leheng Li, Yuning Wang, **Haotian Lin**, Hao Cheng, Zhizhe Liu, Lei He, Jianqiang Wang. "Controllable traffic simulation through llm-guided hierarchical chain-of-thought reasoning". 2025 IEEE/RSJ International Conference on Intelligent Robots and Systems.

SKILLS & FOCUS

- **Reinforcement learning, Robot Foundation Models.**
- Coding skills: Python (PyTorch, JAX), C++, Matlab.
- Robotic Physical Simulator: IsaacSim, IsaacGym, Mujoco
- Software: SolidWorks

EXTRACURRICULAR EXPERIENCES

Secondary Award in Xindong Project (Intelligent vehicle design contest, Tsinghua University)

1/2021

Member of Tsinghua Photography Team (official art group in Tsinghua University)

9/2021 – 6/2024

Committee Member of Tsinghua Photography Team

9/2022 – 6/2023

Member of Student Union (Department Of Art) of Xingjian College

9/2021 – 6/2022